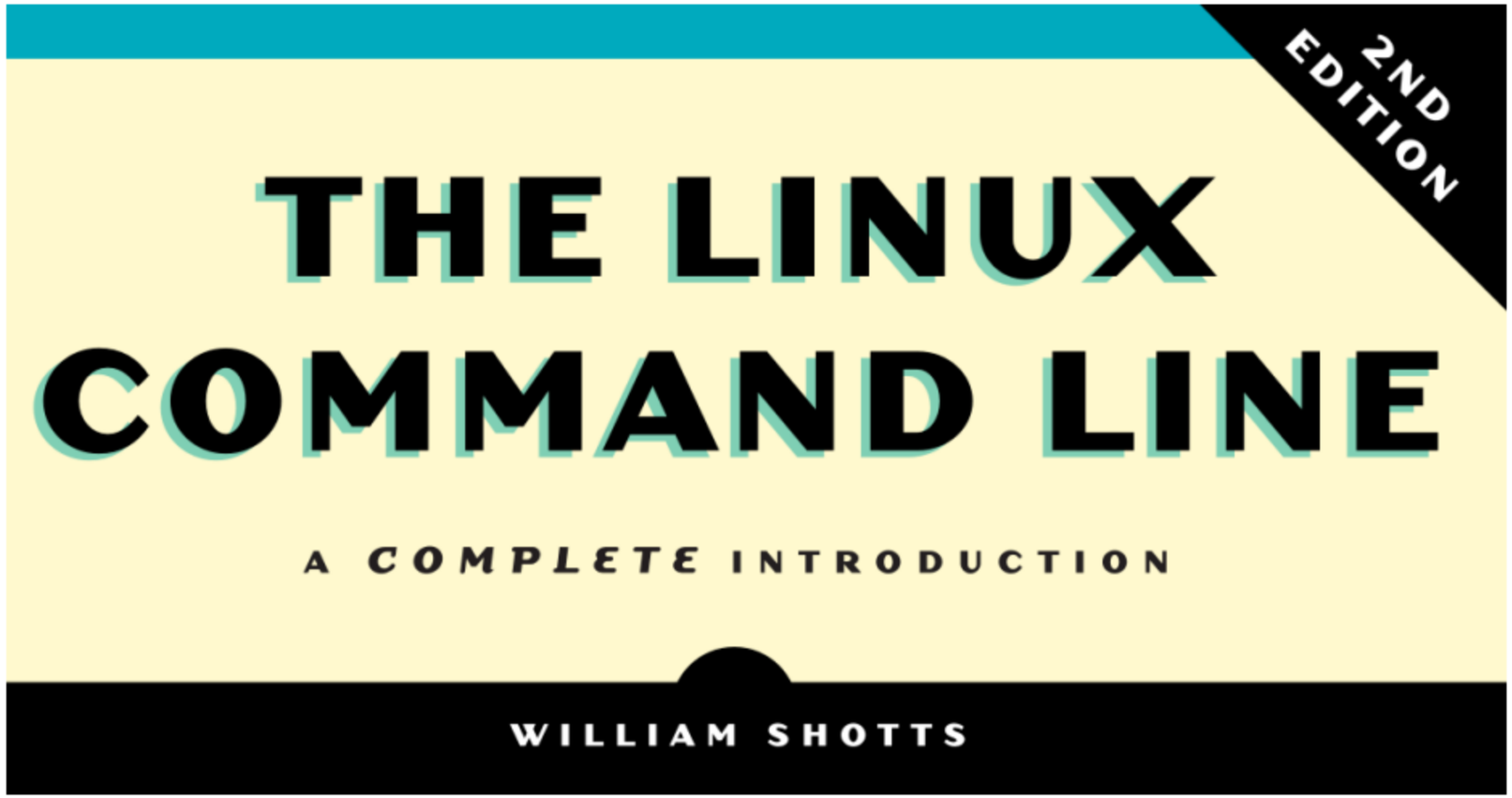
Ubuntu Operating System



https://linuxcommand.org/tlcl.php

Ubuntu OS

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The system can be used

- By multiple users (Multiuser)

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Assuming that you have logged into a Ubuntu System ...

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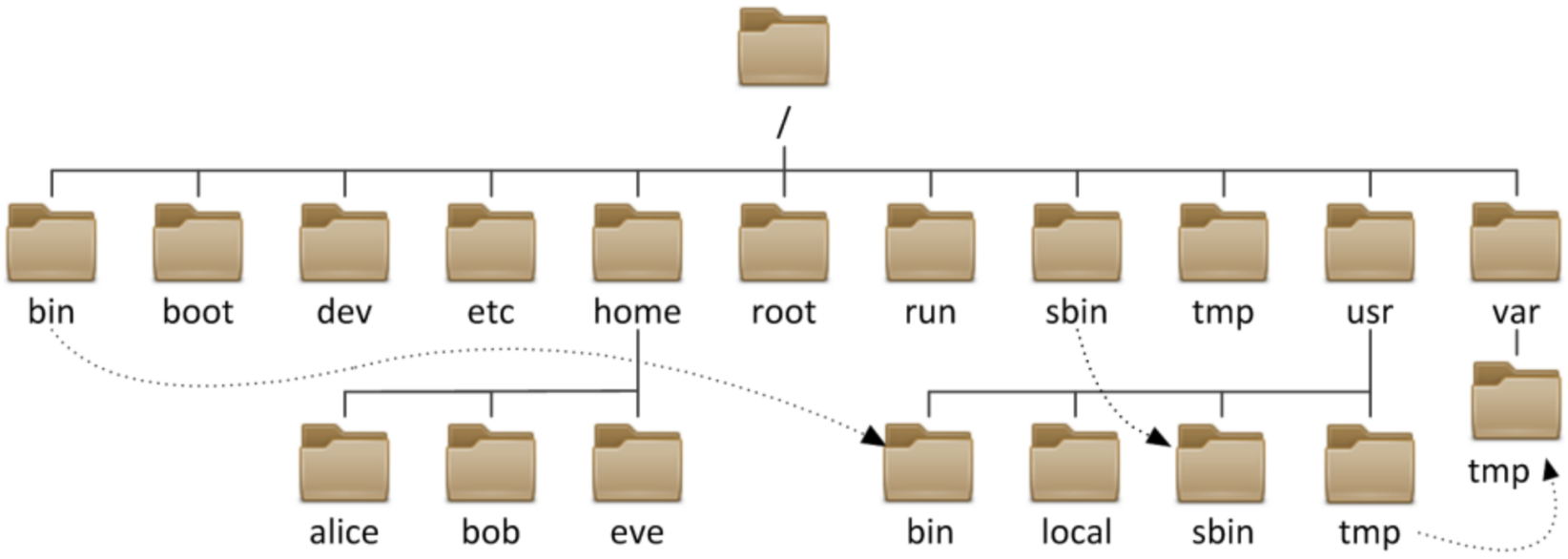
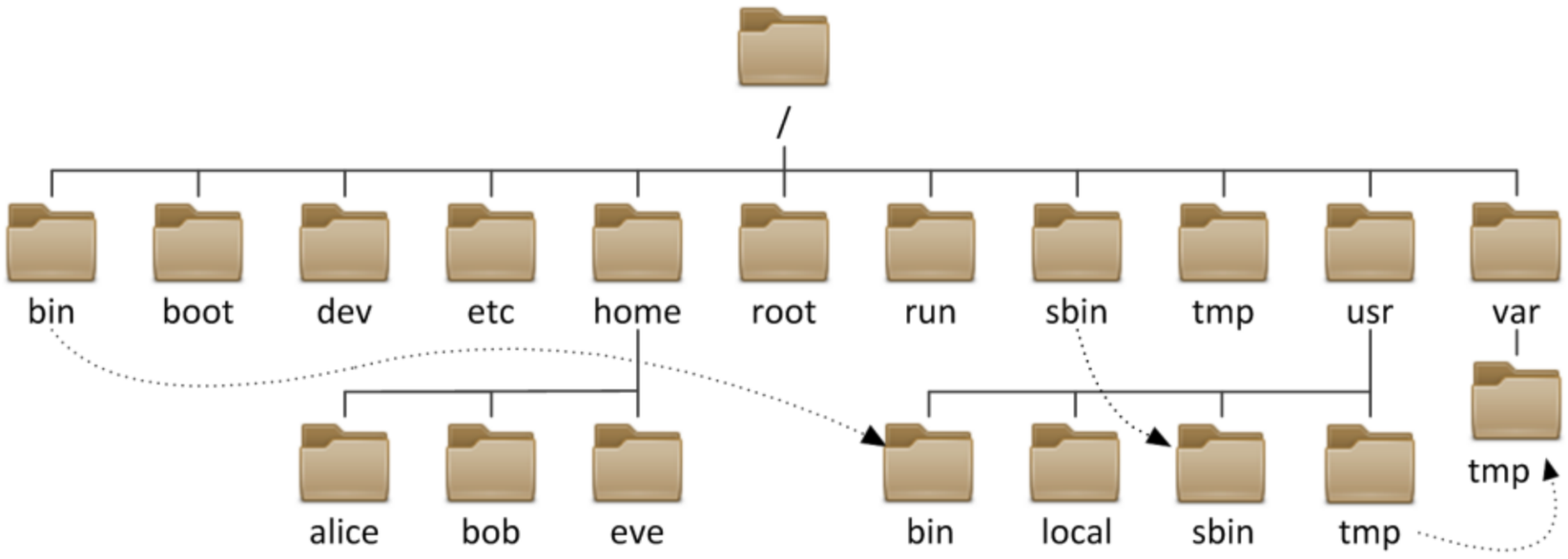
Assuming that you have logged into a Ubuntu System ...

- You will see desktop similar to Windows

Rooted-Tree File System

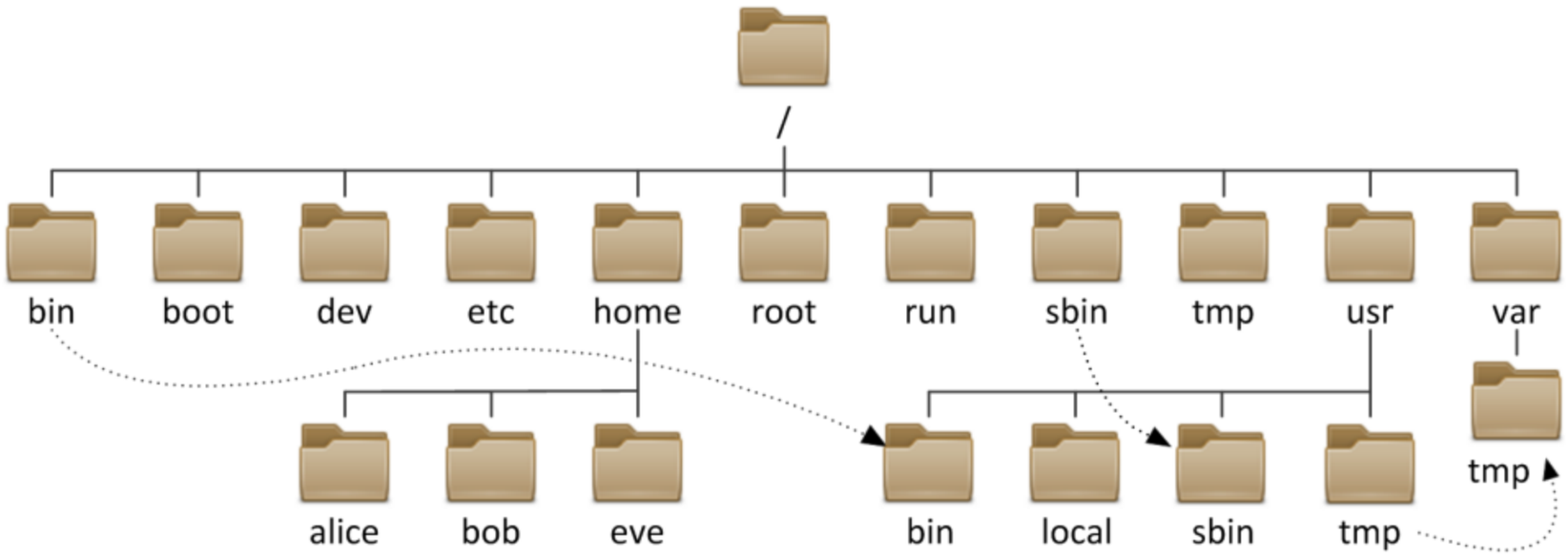


Rooted-Tree File System



As a user, you have a HOME folder (eg. /home/kavita)

Rooted-Tree File System



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Open a `Terminal` from the menu

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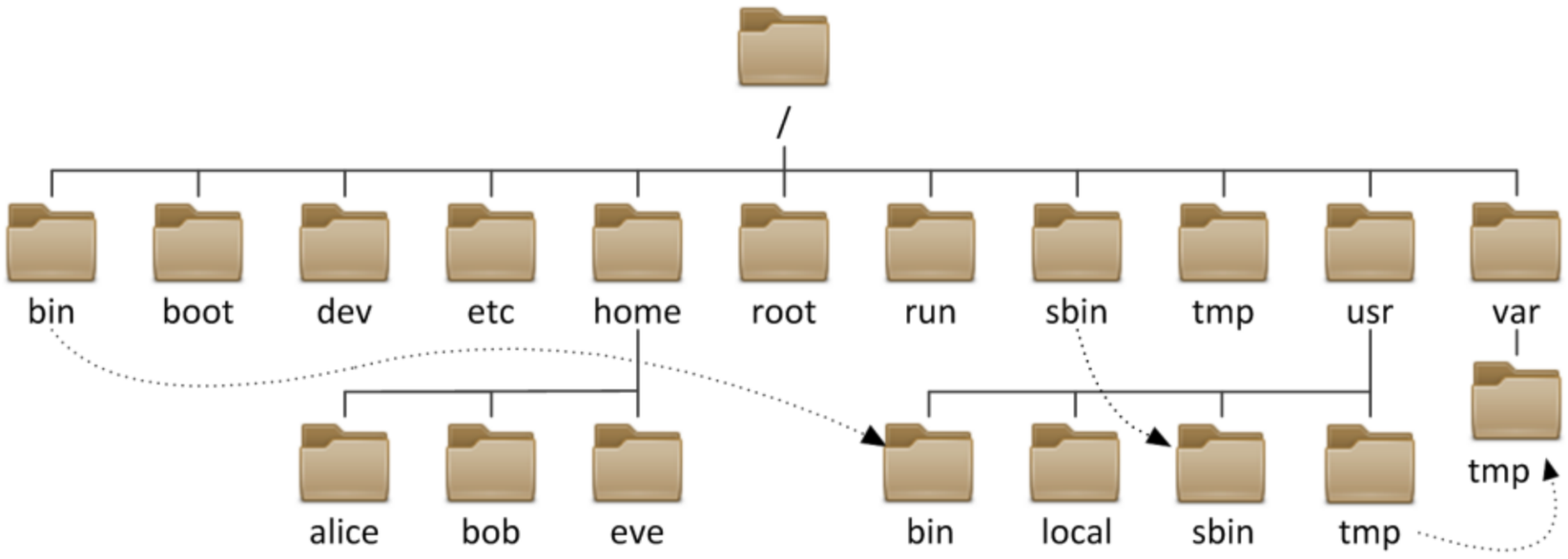

As a user, you have a HOME folder (eg. /home/kavita)

Open a `Terminal` from the menu

Change your password – Two ways –

(a) GUI

(b) Terminal - command `passwd`

HOME folder and Present Working Directory

When you start `Terminal` you are in HOME folder.

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Check the Present Working Directory

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```
$ pwd
/home/...
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Change directory to root

HOME folder and Present Working Directory

When you start `Terminal` you are in HOME folder.

Check the Present Working Directory

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$ pwd
/home/...
```

Change directory to root

```
$ cd /
```

Nothing will be displayed. You can verify using `pwd`.

File Management with System Utilities

List Files

List files in the current folder.

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```
$ ls
bin boot ...
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Linux commands can usually take some parameters.
Option `-l` – show details

```
$ ls -l
total 2.1G
drwxrwxrwt 24 root root 4.0K Oct 8 15:58 tmp
...
```

Shell Commands . . .

Change directory to etc folder:

Shell Commands . . .

Change directory to etc folder:

```
$ cd etc
```

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Change directory to etc folder:

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List files

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Shell Commands . . .

Change directory to etc folder:

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List files

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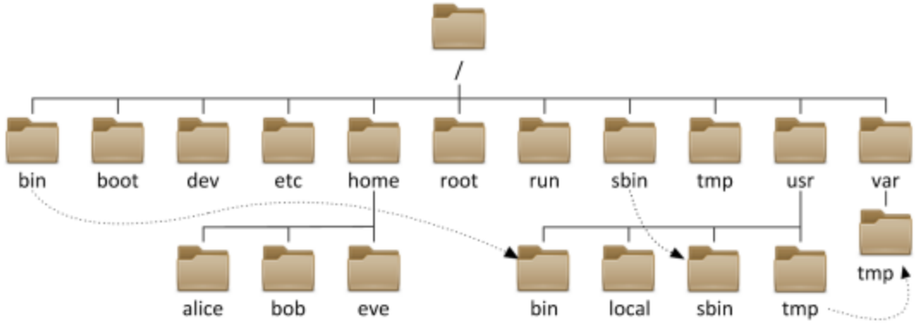
Go to HOME folder

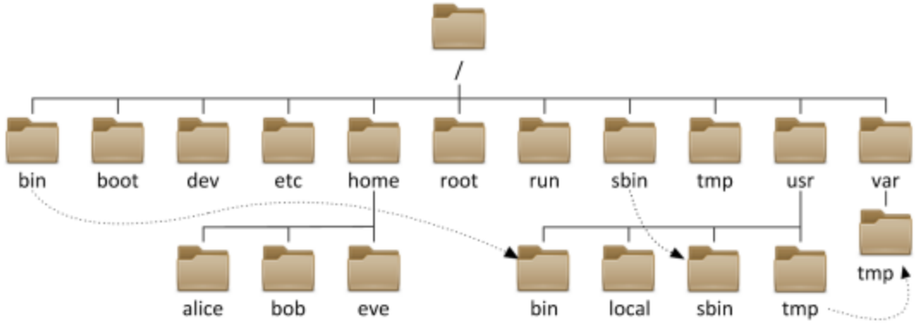
```
$ cd
```

Paths

Paths

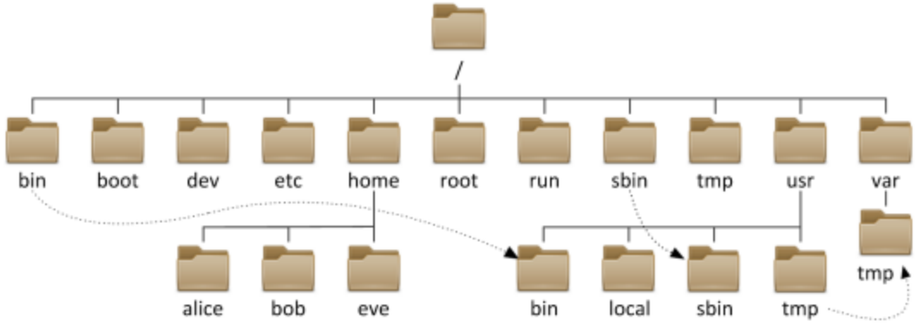

/
bin
boot
dev
etc
home
root
run
sbin
tmp
usr
var
alice
bob
eve
bin
local
sbin
tmp
tmp

Paths



Relative path to a file or folder

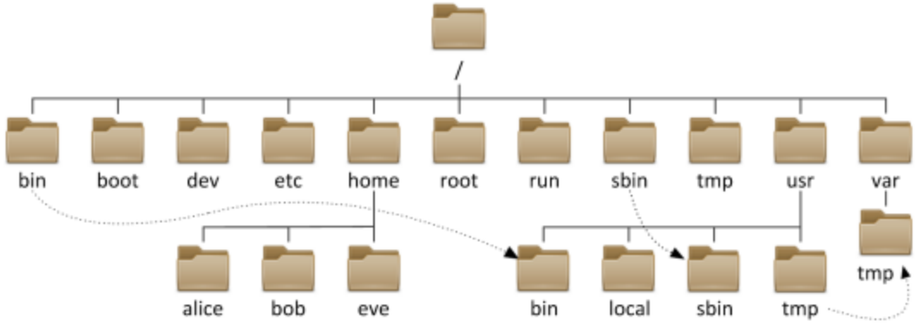
Paths



Relative path to a file or folder

```
$ ls ../../etc
```

Paths



Relative path to a file or folder

```
$ ls ../../etc
```

Absolute path

```
$ ls /etc
```

File Commands . . .

File Commands . . .

```
$ mkdir ds3294
```

Check your current folder and confirm

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Other Commands:

- touch, cat, less # File creation/viewing

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Most commands can take options, eg. `ls -l`

File Commands . . .

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Most commands can take options, eg. `ls -l`

How to find details of a command? `man`

System Info Commands

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CPU Information

System Info Commands

CPU Information

```
$ lscpu
Architecture:  x86_64
CPU op-mode(s):  32-bit, 64-bit
...
Vendor ID: GenuineIntel
Model name:  Intel(R) Core(TM) Ultra 5 125U
Thread(s) per core:  2
Core(s) per socket:  12
Socket(s):  1
...
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Get amount of free and used memory

System Info Commands

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```

System Information ...

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Kernel Info

```
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```

System Information . . .

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Full hardware summary

```
$ lshw -short
```

System Information . . .

Kernel Info

```
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```

Full hardware summary

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```

Block devices

```
$ lsblk
```

Process Information

Get live information about current processes

```
$ top
```

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```
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```

Process Information (Snapshot)

```
$ ps -af
```

Explore System Boot Process

Kernel buffer messages)

```
$ dmesg | less
```

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```
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System boot logs

```
$ journalctl -b
```

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Currently Running Services

```
$ systemctl list-units --type=service --state=running
```